Software Requirements Specification

Student Academic Record Management System (SARMS)

Version 0.5

Prepared By Group 9

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Revision History

VERSION	DATE	DESCRIPTION
0.1	29/01/2026	Created initial SRS structure and analyzed the problem statement
0.2	31/01/2026	Drafted Introduction, Purpose, Scope, and Overall Description sections
0.3	02/02/2026	Defined Functional Requirements and System Features based on SARMS
0.4	04/02/2026	Added Non-Functional Requirements, Interfaces, and Constraints
0.5	06/02/2026	Team review, formatting improvements, and final refinements for submission

3. Introduction

The Student Academic Record Management System (SARMS) is a software tool built to make managing school/college records and office work easier. This document outlines the main plan for the project. It explains how the system will look and what it will cover. By matching what users need with the right technology, this introduction sets the stage for building the software. It ensures the final program solves the hard parts of keeping track of student grades and files.

3.1 Purpose

The main goal of this document is to clearly list what the SARMS software needs to do. It acts as a guide for the developers, testers, and everyone else working on the project. By describing exactly how the system should behave and what rules it must follow, this paper removes any confusion. This ensures that the final software matches exactly what the school needs for its work.

3.2 Document Conventions

To make this document easy to read, we use the following short terms and rules:

- **FAQ:** Stands for **Frequently Asked Questions**. This refers to common questions users ask about the software.
- **SARMS:** Short for **Student Academic Record Management System**. This is the software project described in this document.
- **SWA:** Stands for **Semester Weighted Average**. This is the formula used to calculate a student's grade performance for one term.
- **System:** Refers specifically to the SARMS application.
- **Shall:** Whenever this word is used, it means the requirement is **mandatory**. The team must include it.
- **Structure:** We use numbered sections to make the document easy to navigate.

3.3 Intended Audience

This document is designed for five main groups:

1. **Team Members:** To keep track of the project goals.
2. **Faculty:** To review the quality and accuracy of the work.
3. **Developers:** To use as a blueprint when building the system.
4. **Quality Assurance:** To verify that the features work as promised.
5. **Maintainers:** To help them fix or change the system later on. *Ideally, readers should have basic knowledge of software design.*

3.4 Additional Information

This SRS focuses exclusively on the requirements analysis phase and does not include system design, database schemas, or implementation details. The document corresponds to Version 0.5 of SARMS and may be updated in future revisions if system requirements evolve.

All requirements described in this document are derived from the approved problem statement and academic guidelines.

3.5 Contact information/SRS team members

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3.6 References

1. Problem Statement – Student Academic Record Management System (SARMS)
2. Lab Assignment Instructions for SRS Submission
3. IEEE Std 830 – Software Requirements Specification
4. Software Engineering Lecture Notes

4. Overall Description

4.1 Product Perspective

The Student Academic Record Management System(SARMS) is a web-based application designed to manage and store student academic records digitally. Allows students to enroll in courses independently. SARMS is a standalone application, but can be integrated with existing databases for student and course information. SARMS follows a client-server architecture, where users interact with the application through a web browser, and the data is stored in a cloud-based database or a server.

4.2 Product Functions

The SARMS main functions include student registration, course creation and management, student course enrollment, marks entry and validation, calculation of Semester Weighted Average(SWA), categorisation of students based on academic performance, generation of semester reports, and maintenance of historical academic records. The system is secured with user authentication.

4.3 User Classes and Characteristics

SARMS is a multi-user system. Each user has specific roles and responsibilities in the system.

- **Administrators -** Manage the students' records, maintain course information, and oversee the overall system operations.
- **Faculty Members -** Manage course enrollments, enter and update student marks, and generate academic reports.
- **Students -** Use the system to register, enrol in courses, and view their academic performance and semester reports.

4.4 Operating Environment

SARMS is a web-based application that will operate in a client-server environment. The application can be accessed through standard web browsers such as Google Chrome, Opera GX, Mozilla Firefox, and Microsoft Edge. The backend of the application will run on a cloud-based server or a physical server. The application is operating system independent and can be run on any operating system.

4.5 User Environment

User interaction with the SARMS is a simple web-based interface. The application is designed to be simple to use and easily accessible, requiring only basic knowledge of computers and navigation skills. Users do not need any special training to navigate the application. Readable text, clear navigation and an optional accessibility feature are included to ensure comfortable usage for any user.

4.6 Design/Implementation Constraints

The SARMS system must be developed as a web-based application using standard web technologies. Must comply with institutional policies, data protection regulations, and applicable web accessibility standards. The application must ensure secure data handling, support role-based access control, and be compatible with major web browsers. All third-party tools or libraries used must comply with open-source or institutional licensing requirements.

4.7 Assumptions and Dependencies

- **Assumptions -** Users have a stable internet connection and a compatible web browser. It is also assumed that the academic data provided to the system is accurate and maintained regularly.
- **Dependencies -** The system depends on the availability of cloud services and database services for proper operation. Any API or service integration with external systems is dependent on the availability and reliability of the corresponding API's and services.

5. External Interface Requirements

This part of the document explains how SARMS works with people, computer equipment, and other software. It also describes the rules the system uses to send and receive data.

5.1 User Interfaces

- The software provides an easy-to-use design for both teachers and administrators.
- Administrators can use the interface to set up new classes and manage student files.
- Teachers can enter grades and create term reports directly through the system.
- The SWA score appears clearly on every student report so it is easy to find.
- The design stays simple and is easy for anyone to move around in.
- The system checks all entries (like names and grades) to help prevent mistakes.

5.2 Hardware Interfaces

- The system works on normal computers, such as desktops and laptops.
- You do not need any special or extra hardware to run the software.
- Users only need a standard keyboard and mouse to enter data.
- The system shows all information on regular computer monitors or screens.

5.3 Software Interfaces

- The system connects to a database to save student files, course info, and grades.
- It can link up with other college data systems if it needs to.
- It pulls information from the database to create neatly organized term reports.
- The software includes all the parts needed to store and manage academic records.

5.4 Communication protocol and Interfaces

- The system works over a normal network or internet connection.
- Different parts of the software send data to each other safely and reliably.
- Information flows smoothly between what the user sees and the background system.
- The system makes sure information doesn't get changed or corrupted while it is being sent.

6. System Features

6.1 Student Registration and Record Creation

6.1.1 Description and Priority

This feature enables the registration of new students to the system by using their personal and academic details. Each student record includes a unique roll number, address, semester number and courses registered for the semester.
This feature is foundational to the system and thus it is assigned high priority.

6.1.2 Action/Result

Action: Administrator requests to register a new student
Response: System displays a registration form requiring student details

Action: Administrator enters student details and submits the student information
Response: system validates input, generates a unique roll number and stores the student record in the database

6.1.3 Functional Requirements

Feature requirements: Student Registration (FR-SR)

Feature ID	Feature Name	Description
FR-SR1	Register Student	System will allow the entry of a student's personal and academic details.
FR-SR2	Generate Roll Number	System will generate a unique roll number for each registered student.
FR-SR3	Validate Input Data	System will validate the required fields before registration.
FR-SR4	Store student Data	System will securely store student records in the database,
FR-SR5	Maintain Semester Records	System will maintain semester-wise academic records for each student.

6.2 Course management

6.2.1 Description and Priority

This feature allows the administrators or the authorised users to create and manage courses within the system. Each course includes course code, title, number of credits and syllabus information.

6.2.2 Action/Result

Action:administrator requests to create a new course
Response: System provides a form to enter course details

Action: Administrator submits course information
Response: system validates the input and stores the course record in the database

6.2.3 Functional Requirements

Feature requirements: course management (FR-CM)

Feature ID	Feature Name	Description
FR-CM1	Create course	System will allow the creation of the course with a unique course number.
FR-CM2	Store Credits	System will store the number of credits for each course.
FR-CM3	Store Syllabus	System will store the syllabus information for each course.
FR-CM4	Modify Course	System will allow modification of course details
FR-CM5	Delete course	System will allow deletion of courses.

6.3 Student course enrollment

6.3.1 Description and Priority

This feature enables admission of students into courses for a semester.
This feature is assigned high priority.

6.3.2 Action/Result

Action: Administrator assigns courses to a student
Response: System records enrollment and prevents duplicate entry
Action: Administrator submits course information
Response: System validates the input and stores the course record in the database

6.3.3 Functional Requirements

Feature requirements: course enrollment (FR-EN)

Feature ID	Feature Name	Description
FR-EN1	Enroll Student	System will allow enrollment of students into available courses.
FR-EN2	Prevent Duplicate Enrollment	System will prevent duplicate enrollment in the same course.
FR-EN3	Maintain Enrollment Records	System will maintain semester-wise enrollment records.

6.4 Marks Entry And Storage

6.4.1 Description and Priority

This feature allows the entry of marks for credited courses taken by the students during a semester.

This feature is assigned high priority.

6.4.2 Action/Result

Action: Authorised user enters marks for each courses taken by a student

Response: System validates and stores marks for the semester

6.4.3 Functional Requirements

Feature requirements: Marks Entry And Storage (FR-ME)

Feature ID	Feature Name	Description
FR-ME1	Enter Marks	System will allow entry of marks for each credited course.
FR-ME2	Validate Marks	System will ensure marks entered are within valid limits.
FR-ME3	Store Semester Marks	System will store marks semester-wise for each student.

FR-ME4	Update Academic Records	System will update the student academic records after marks are entered.

6.5 Semester Weighted Average (SWA) Calculation

6.5.1 Description and Priority

This feature calculates the Semester Weighted Average (SWA) using marks and course credit points for the current semester.
This feature is assigned high priority.

6.5.2 Action/Result

Action: Marks are finalised for the semester.
Response: System calculates SWA using credit-weighted formula and stores the result.

6.5.3 Functional Requirements

Feature requirements: SWA Calculation (FR-SWA)

Feature ID	**Feature Name**	**Description**
FR-SWA1	Calculate Semester SWA	System will calculate SWA using credits and marks for the current semester.
FR-SWA2	Store Semester SWA	System will store the calculated SWA for the semester.
FR-SWA3	Display Semester SWA	System will display SWA in the student record.

6.6 Cumulative Weighted Average and Academic Standing

6.6.1 Description and Priority

This feature is used to calculate cumulative weighted average across semesters and determines academic standing based on SWA thresholds. This feature is assigned high priority.

6.6.2 Action/Result

Action: Cumulative Weighted average of all semesters is computed.

Criteria for VC list, conditional setting is set.

Response: System updates cumulative weighted average and determines academic standing.

6.6.3 Functional Requirements

Feature requirements: Academic Standing (FR-AS)

Feature ID	**Feature Name**	**Description**
FR-AS1	Calculate Cumulative Average	System will compute cumulative weighted average across semesters using credit points.
FR-AS2	Determine VC List	System will place students with SWA>=85 on the Vice Chancellor's list.
FR-AS3	Determine Conditional Standing	System will place students with SWA<=50 on Conditional Standing.
FR-AS4	Store Academic Status	System will store and display the student's academic standing.

6.7 Semester Report Generation

6.7.1 Description and Priority

This feature generated formatted semester formatted semester reports containing marks, SWA, cumulative average and academic standing.

This feature is assigned medium priority.

6.7.2 Action/Result

Action: User requires generation of semester report

Response: System compiles academic data and generates a formatted printable report

6.7.3 Functional Requirements

Feature requirements: Report Generation (FR-RG)

Feature ID	**Feature Name**	**Description**
FR-RG1	Generate Report	System will generate formatted semester reports.
FR-RG2	Display SWA	System will display SWA on the report.
FR-RG3	Display Cumulative Average	System will display cumulative weighted average on the report.
FR-RG4	Display Academic Standing	System will display academic standing on the report.
FR-RG5	Print Report	System will allow printing of the semester report.

7. Other Non-Functional Requirements

This part of the document explains the non-functional requirements for SARMS. Instead of listing new features, this section defines *how* the system should perform. It sets the rules for speed, security, and reliability to make sure student records are managed safely and without errors.

7.1 Performance requirements

1**.** The application is expected to manage reports and marks without keeping the user waiting.
2. Calculations for the SWA happen automatically as soon as the data is entered.
3. Many users can access the system at once without slowing it down.
4. Retrieving a student's record takes very little time under normal conditions.
5. The system keeps running smoothly even as the number of student records increases.

7.2 Safety requirements

1. Protect student records to stop them from being lost or damaged by accident.
2. Keep data accurate and consistent while teachers enter marks or the system calculates averages.
3. Block anyone from deleting or changing academic files without the right permission.
4. Have a backup plan ready so data can be restored if the system crashes.
5. Note that this system is software-only and does not control any dangerous machinery or hardware.

7.3 Security requirements

1**.** Require every user to authenticate (log in) before accessing any feature.
2. Restrict access to sensitive details so users only see what their job allows.
3. Allow only authorized personnel to input or edit academic marks.
4. Protect student history and grade calculations from being viewed by the wrong people.
5. Ensure all data is secure while it is being saved to the database or transferred.

7.4 Software quality attributes

1. *Reliability*: The system shall perform its intended functions consistently without data errors.
2. *Availability*: The system shall be available during normal academic and administrative hours.
3. *Usability*: The system shall be easy to use and understandable for administrators and faculty.
4. *Maintainability:* The system shall be designed to support future updates and enhancements.
5. *Scalability*: The system shall be capable of handling an increasing number of students and courses.

7.5 Project documentation

1**.** Create proper documentation to explain how the system works and what it requires.
2. Use this SRS document as the main guide for the entire project.
3. Keep all documents updated to help with future repairs or improvements.
4. Write everything in clear, simple language so it is easy to read.

7.6 User documentation

1. Provide basic guides to help administrators and teachers use the software.
2. Include instructions for key tasks like setting up courses, enrolling students, entering grades, and printing reports.
3. Make sure all manuals are easy to read and easy to find.
4. Offer built-in help tips to assist users with common daily tasks.